

1-6-1: Verification and Documentation in Troubleshooting

At the end of this episode, I will be able to:

1. Identify and explain the importance of verifying system functionality and documentation in troubleshooting, given a scenario.

Learner Objective: *Identify and explain the importance of verifying system functionality and documentation in troubleshooting, given a scenario.*

Description: In this episode, the learner will examine the verification and documentation steps when applying a troubleshooting methodology. We will explore the techniques and examples of these steps in networks.

- When we are trying to verify that are solution did indeed work?
 - o Start with a confirmation of resolution, post-implementation
 - ♣ Confirm that the original problem has been resolved by retesting under the same conditions that initially identified the issue.
 - ♣ Engage with the end-users or stakeholders to ensure the solution meets the needs, resolving the issue from their perspective.
 - o Perform system-wide and platform-wide checks
 - ♣ Perform a comprehensive system checks, ensuring the implemented solution has not adversely affected other areas of the network.
 - ♣ Monitor system performance and logs to detect any unintended consequences of the changes made.
- If we have confirmed that our implementation has solved the problem, what is next?
 - o Provide final documentation (may be located in more than a single location)
 - ♣ Update network documentation to reflect any changes made during the troubleshooting process.
 - ♣ Document the problem, the analysis, the implemented solution, and the verification process for future reference and learning.
- Can you provide a few examples of these steps?
 - o Examples
 - ♣ Switch Reconfiguration
 - ♣ Perform confirmation testing
 - ♣ After reconfiguring the switch, conduct tests to ensure network stability and VoIP quality.
 - ♣ System-wide Checks
 - ♣ Verify that the new VLAN settings and prioritization rules have not negatively impacted other network traffic.
 - ♣ Final Documentation
 - ♣ Record the details of the switch reconfiguration
 - ♣ Configuration changes
 - ♣ The reasoning or justification behind the changes
 - ♣ The results of post-implementation testing
 - ♣ Virtualization host reconfiguration
 - ♣ Perform confirmation testing
 - ♣ Test connectivity and network performance for all VMs running on the host
 - ♣ Ensure the VMs have not been adversely affected by the changes.
 - ♣ Perform system-wide and platform-wide checks
 - ♣

- Perform continuous monitoring on the host and other VMs to identify any unintended side effects of the network adapter reconfiguration.

- ♣ Final Documentation

- ♣ Record the changes made to the virtual network adapter settings
- ♣ Record the reasoning or justification for the change
- ♣ Record the outcomes of the verification process.

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- Additional Resources:
 - o If applicable